Acids & Bases - Questions

1. Acids & Bases

1. Identify each of the following substances as a strong acid, a weak acid, a strong base or a weak base:

nitric acid	ammonia	$Ca(OH)_2$
sodium carbonate	H_2SO_4	phosphoric acid
potassium hydroxide	vinegar	sodium hydrogencarbonate
HCl	KOH	citric acid

2. Complete the gaps:

The theory that an acid is a substance which produces h…… ions in water was first proposed by the Swedish chemist, Svante Arrhenius. Arrhenius proposed that hydrogen ions were produced by the ionisation of an a…….. in water. For example, in hydrochloric acid, the hydrogen chloride molecules are ionised to form h…… ions and c…….. ions: HCl → ……. + ……

Similarly, sulfuric acid and nitric acid i…….. in aqueous solutions to form h…….. ions. Each of these acids is a s……. acid because it is essentially completely ionised in aqueous solution. In hydrochloric acid, for example, there are very few unionised h…….. c……. molecules. Virtually a…….. the molecules have been ionised to form ……. and ……..ions.

In many acids such as e…….. acid and citric acid only a small proportion of the dissolved molecules are ionised. These acids are called w…… acids. The equation for the ionisation of acetic acid is as follows: CH_3COOH → …… + ……

A solution of acetic acid would therefore contain …… , ……. and un-ionised ……., together with water.

3. Give the formulae of the particles, apart from water, that would be present in the following solutions. Choose your answers from the formulae given in the box:

H^+	OH^-	HCl	SO_4^{2-}	CH_3COO^-	K^+	NO_3^-
Br^-	Cl^-	CO_3^{2-}	CH_3COOH	HNO_3	Na^+	

a. a solution of hydrochloric acid
b. a solution of nitric acid
c. a solution of ethanoic acid
d. a solution of sodium hydroxide

4. Give the colour to which you would expect the indicator to change, when added to each of the following aqueous solutions:

Solution	Indicator	
	Phenolphthalein	Litmus
nitric acid		
calcium hydroxide		
ammonia		
H_2SO_4		

5. Identify each of the following statements as true or false. If the statement is false, then rewrite it to make it true.

   a. Acetic acid and citric acid are examples of weak acids.

   b. Hydrochloric acid and sodium hydroxide are examples of strong acids.

   c. When an acid dissolves in water, many hydroxide ions are formed.

   d. Nitric acid turns litmus a red colour.

   e. A potassium hydroxide solution will turn litmus a blue colour and phenolphthalein a pink colour.

   f. When hydrogen chloride (HCl) is added to water, only some of the dissolved molecules break up into H^+ and Cl^- ions.

   g. When ammonia is added to water, only some of the dissolved ammonia molecules form OH^- ions.

   h. Carbonic acid is present in "fizzy" drinks.

   i. Orange juice has a pH greater than 7.

   j. Ammonia solution has a pH greater than 7.

   k. A neutral solution has a pH of 7.0.

   l. All acidic solution have pH values less than 7.0.

   m. Sodium carbonate and sodium hydrogencarbonate are examples of weak acids.

6. Match up the correct pH value with the corresponding solution:

a concentrated solution of hydrochloric acid	pH 7.0
a solution of ammonia	pH 1.0
a concentrated solution of sodium hydroxide	pH 3.5
pure water	pH 10.0
vinegar	pH 14.0

7. For the following questions, choose the best alternative:

   a. Acetic acid is a weak acid because

   A it is not very soluble in water
   B it is not a very good conductor of electricity
   C it forms only a few hydrogen ions in solution
   D it is produced by living organisms

   b. Which solutions will both turn methyl orange a red colour?

   A sodium hydroxide and ammonia
   B potassium hydroxide and oxalic acid
   C soda water and hydrochloric acid
   D lemon juice and sodium carbonate

   c. Which of the following properties is NOT typical of a base?

   A when dissolved in water, it forms hydroxide ions
   B turns phenolphthalein pink
   C when dissolved in water, it forms a solution with a pH greater than 7
   D it forms hydrogen ions when dissolved in water

   d. Nitrous acid is a weak acid and it has the formula of HNO_2. Apart from water molecules, which particles would you expect to be present in a solution of HNO_2?

   A OH^- and NO B only HNO_2 C H^+ and NO_2^- D H^+, NO_2^- and HNO_2

   e. Hydrobromic acid is a strong acid and it has the formula of HBr. Apart from water molecules, which particles would you expect to be present in a solution of hydrobromic acid?

   A OH^- and Br^- B only HBr C H^+ and Br^- D H^+, Br^- and HBr

   f. Barium hydroxide has the formula of $Ba(OH)_2$. Apart from water molecules, which particles would you expect to be present in a solution of barium hydroxide?

   A OH^- and Ba^{2+} B only $Ba(OH)_2$ C H^+ and BaO D OH^-, Ba^{2+} and $Ba(OH)_2$

2. Reactions of Acids.

8. Complete the following equations by supplying the formulae of the missing substances:

   a. hydrochloric acid + sodium oxide → .....…… + salt solution

   b. nitric acid + potassium hydroxide → .....…….. + salt solution

   c. hydrochloric acid + zinc → …… + salt solution

   d. sodium carbonate + hydrochloric acid → .....……. + ...…… + salt solution

   e. sulfuric acid + sodium hydroxide → .....…….. + salt solution

   f. magnesium carbonate + nitric acid → .....…….. + ...…….+ salt solution

   g. sulfuric acid + magnesium → ..... + salt solution

   h. nitric acid + magnesium oxide → .....…… + salt solution

9. A salt is an ionic compound, that is, it is made up of a positive ion and a negative ion.
   Salts are formed in acid reactions.
   The negative ion of the salt comes from the acid and the positive ion from the other reactant.

   Hydrochloric acid ($H^+ + Cl^-$) forms salts containing the chloride ion (Cl^-)
   Nitric acid ($H^+ + NO_3^-$) forms salts containing the nitrate ion (NO_3^{--})
   Sulfuric acid ($H^+ + SO_4^{2-}$) forms salts containing the sulfate ion (SO_4^{2-})

   Give the products (names or formulae) that are formed in the following reactions:

   a. hydrochloric acid + sodium hydroxide solution → ..... + solution of .....

   b. nitric acid + calcium oxide solution → ..... + solution of .....

   c. sulfuric acid + copper carbonate solution → ..... + ... + solution of .....

   d. hydrochloric acid + magnesium → ..... + solution of .....

10. Write an ionic equation for the reactions that occur for each of the following

    a. hydrochloric acid solution is added to solid sodium oxide

    b. dilute hydrochloric acid solution is added to a solution of potassium carbonate

    c. hydrochloric acid solution is added to a solution of sodium hydroxide

    d. sulfuric acid solution is added to solid copper hydroxide

    e. dilute hydrochloric acid is added to zinc

    f. dilute sulfuric acid is added to magnesium

    g. sulfuric acid solution is added to sodium carbonate solid

    h. dilute sulfuric acid is added to calcium

    i. solid magnesium carbonate is added to dilute hydrochloric acid

    j. nitric acid is added to solid zinc oxide

11. Describe what you would expect to see happening during the reactions

a. dilute hydrochloric acid solution is added to a solution of potassium carbonate

b. sulfuric acid solution is added to solid copper hydroxide

c. nitric acid solution is added to some magnesium

12. Some properties of substances A, B, C, D, E and F are described in the table below.
Identify each of these substances from the list given in the box:

copper carbonate	sodium carbonate	zinc
copper oxide	magnesium hydroxide	calcium oxide

Substance	Appearance	Solubility in water	Reaction with dilute hydrochloric acid
A	white solid	insoluble	dissolves to form a colourless solution
B	green solid	insoluble	reacts to form a blue solution and a colourless gas that turns limewater milky
C	white solid	soluble	dissolves to form a colourless solution
D	silver solid	insoluble	reacts to form a gas that "pops" when ignited
E	black solid	insoluble	reacts slowly to form a blue solution
F	white solid	soluble	reacts to form a colourless solution and a gas that turns limewater milky

13 Three bottles containing solutions have 'lost' their labels. However, it is known that one of the solutions is sodium chloride, another is sodium carbonate and the third is copper nitrate . Describe how you could distinguish between each of the three solutions.

14. Fill in the gaps

Names of reacting substances	Name of salt formed in the reaction	Names of other products formed
magnesium oxide + nitric acid	……. ……..	……..
zinc + hydrochloric acid	……. ……..	……
sodium carbonate + ……. acid	sodium chloride	…….. + …….
……..…. hydroxide + …… acid	sodium sulfate	……
potassium …….. + nitric acid	……. ……..	carbon dioxide + water
……. + sulfuric acid	magnesium …….	hydrogen

15. Describe what you would observe for each of the following experiments:

a. a drop of hydrochloric acid solution is put on a piece of blue litmus paper.

b. several drops of phenolphthalein solution are added to a solution of calcium hydroxide.

c. some carbon dioxide is bubbled through some limewater

d. a burning match is placed at the top of a test tube that contains hydrogen

e. some sulfuric acid solution is added to some sodium carbonate.

f. a small piece of magnesium is added to a solution of hydrochloric acid.

g. a solution of copper sulfate is heated until all the liquid evaporates.

16. Write balanced ionic equations for the following:

a. dilute nitric acid is added to a solution of sodium hydroxide

b. solid potassium carbonate is added to a solution of hydrochloric acid.

c. a solution of sodium carbonate is added to a solution of nitric acid

d. dilute sulfuric acid is added to calcium

e. dilute hydrochloric acid is added to some solid potassium oxide.

17. Identify each of the substances A - I by choosing a name from the following list:

copper	copper carbonate	potassium carbonate
zinc	dilute hydrochloric acid	soda water
a solution of ammonia	a solution of potassium hydroxide	sodium hydroxide

Substance A - a white solid that produces a colourless gas when added to dilute hydrochloric acid. The colourless gas turns limewater milky. A solution of A has a pH of about 8.5.

Substance B - a silver-coloured solid that reacts with dilute hydrochloric acid to form a gas which "pops" when ignited.

Substance C - a colourless solution which turns litmus red, but is a very poor conductor of electricity.

Substance D - a white solid which forms only a colourless solution when added to dilute nitric acid.

Substance E - a colourless solution which smells like floor cleaner. It turns phenolphthalein pink.

Substance F - a green solid which produce a colourless gas when added to dilute nitric acid

Substance G - a brown-coloured shiny solid which does not react with acid.

Substance H - a colourless solution which turns litmus red and is a very good conductor of electricity.

Substance I - a colourless solution that turns litmus blue and is a very good electrical conductor.

18. Complete the following equations by supplying the formulae of the missing substances:

a. hydrochloric acid + zinc oxide → .....…… + salt solution

b. nitric acid + potassium hydrogencarbonate → .....……. + ...…… + salt solution

c. hydrochloric acid + magnesium → …… + salt solution

d. sodium sulfite + sulfuric acid → .....……. + ...…… + salt solution

e. nitric acid + sodium sulfide → .....…….. + salt solution

f. magnesium carbonate + nitric acid → .....……. + ...…… + salt solution

g. sulfuric acid + potassium sulfite → .....……. + ...…… + salt solution

h. hydrochloric acid + magnesium hydrogencarbonate → .....……. + ...…… + salt solution

19. Give the products (names or formulae) that are formed in the following reactions:

a. hydrochloric acid + potassium hydroxide → ..... + solution of .....

b. calcium sulfite + nitric acid → .....……. + ...…… + solution of .....

c. sulfuric acid + magnesium sulfide → .....…….. + solution of .....

d. sodium hydrogencarbonate + nitric acid → .....… + ...…… + solution of .....

e. hydrochloric acid + sodium sulfite → ..... + ...…… + solution of .....

f. hydrochloric acid + magnesium carbonate → .....…… + ...…… + solution of .....

g. nitric acid + magnesium oxide → ..... + solution of .....

h. sulfuric acid + potassium sulfide→ ..... + solution of .....

i. hydrochloric acid + barium → ..... + solution of .....

20. Write an ionic equation for each of the following reactions:

a. hydrochloric acid solution is added to solid magnesium oxide

b. dilute nitric acid solution is added to a solution of potassium sulfite

c. sulfuric acid solution is added to a solution of sodium hydrgencarbonate

d. hydrochloric acid solution is added to solid copper sulfide

e. dilute hydrochloric acid is added to magnesium

f. dilute sulfuric acid is added to potassium hydroxide solution

g. sulfuric acid solution is added to sodium sulfide solid

h. dilute sulfuric acid is added to potassium

i. solid magnesium sulfite is added to dilute hydrochloric acid

j. nitric acid solution is added to potassium sulfide solution

21. Describe what you would expect to see happening during the reactions

    a. dilute hydrochloric acid solution is added to a solution of sodium hydrogencarbonate

    b. sulfuric acid solution is added to solid copper sulfide

    c. nitric acid solution is added to some magnesium sulfite